

## Research Paper

# A common-normal-based framework for efficient ellipse contact detection in discrete element modelling

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## ABSTRACT

Contact modelling for ellipses/ellipsoids within the discrete element framework has long been a challenging area. Currently, four methods, namely the intersection method, geometric potential method, midway method, and common-normal method, have been proposed to define contact features between ellipses. This study addresses two key objectives: first, to evaluate whether any of these methods are energy-conserving, and second, to propose an accurate, stable and efficient contact detection framework based on the common-normal method for 2D problems. In this framework, the Gilbert–Johnson–Keerthi (GJK) algorithm is used for contact detection between ellipses, while a hybrid expanding polytope algorithm (EPA)–Levenberg–Marquardt (LM) approach is employed to compute contact features. The proposed framework is verified through two numerical examples focusing on energy conservation. Its robustness and applicability are demonstrated by modelling an illustration example involving a large number of ellipses. Detailed computational performance and advantages of the proposed framework are also provided.

## 1. Introduction

The discrete element method (DEM) (Cundall and Strack, 1979; Feng, 2021a, b) has been widely used to simulate geotechnical engineering problems involving complex particulate features (Qu et al., 2019, 2022; Zeng et al., 2025). Over the past decade, particle shapes in DEM simulations have transitioned from circular/spherical to non-circular/spherical forms, reflecting the real particle shapes in nature. Among non-spherical particle shapes, ellipses/ellipsoids are considered the simplest, and have gained more attention since the 1990s (Rothenburg and Bathurst, 1992; Zhao and Zhao, 2019; Jiang et al., 2024).

To accurately compute contact force between contact pairs of ellipses/ellipsoids, it is essential to clearly define contact features. To date, four approaches have been proposed: the intersection method (Rothenburg and Bathurst, 1991; Ting et al., 1992), the geometric potential method (Ng, 1994; Lin and Ng, 1995), the midway method (Houlsby, 2009), and the common-normal method (Ng, 1994). Detailed

descriptions and evolutions of these methods are provided in Feng (2023) and Kheradmand et al. (2022). However, due to the differing definitions within each method, they gave varying contact features (Lin and Ng, 1995). A fundamental question arises: are any of these methods energy-conserving? Addressing this question is the primary objective of this paper.

From the computational efficiency point of view, the numerical performance (accuracy, stability, and efficiency) of solving strategies is crucial in selecting a contact detection method (Lin and Ng, 1995). Since all four methods are described by sets of nonlinear equations, Newton-based iteration strategies are commonly used to solve them numerically (Dziugys and Peters, 2001; Podlozhnyuk et al., 2016; Kheradmand et al., 2020). To enhance its robustness, various techniques were proposed to provide a better initial guess. Meanwhile, Wellmann et al. (2008) introduced a common-normal concept that simplifies the 3D problem with eight variables into a 2D global optimisation problem with two unknowns for the common-normal method. They further proposed a more effective solution by combining Newton's method and Levenberg-

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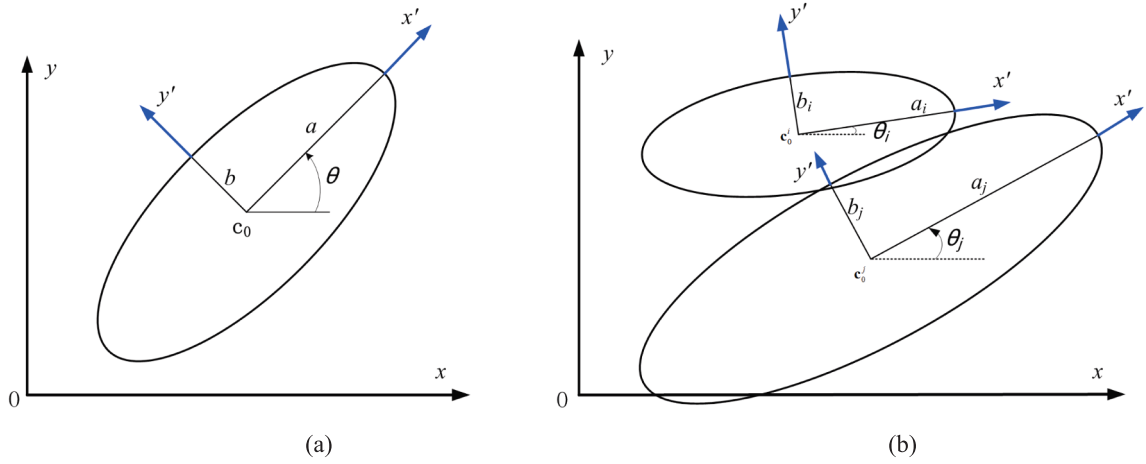


Fig. 1. Ellipse in global and local coordinate systems: (a) one ellipse; (b) two contacting ellipses.

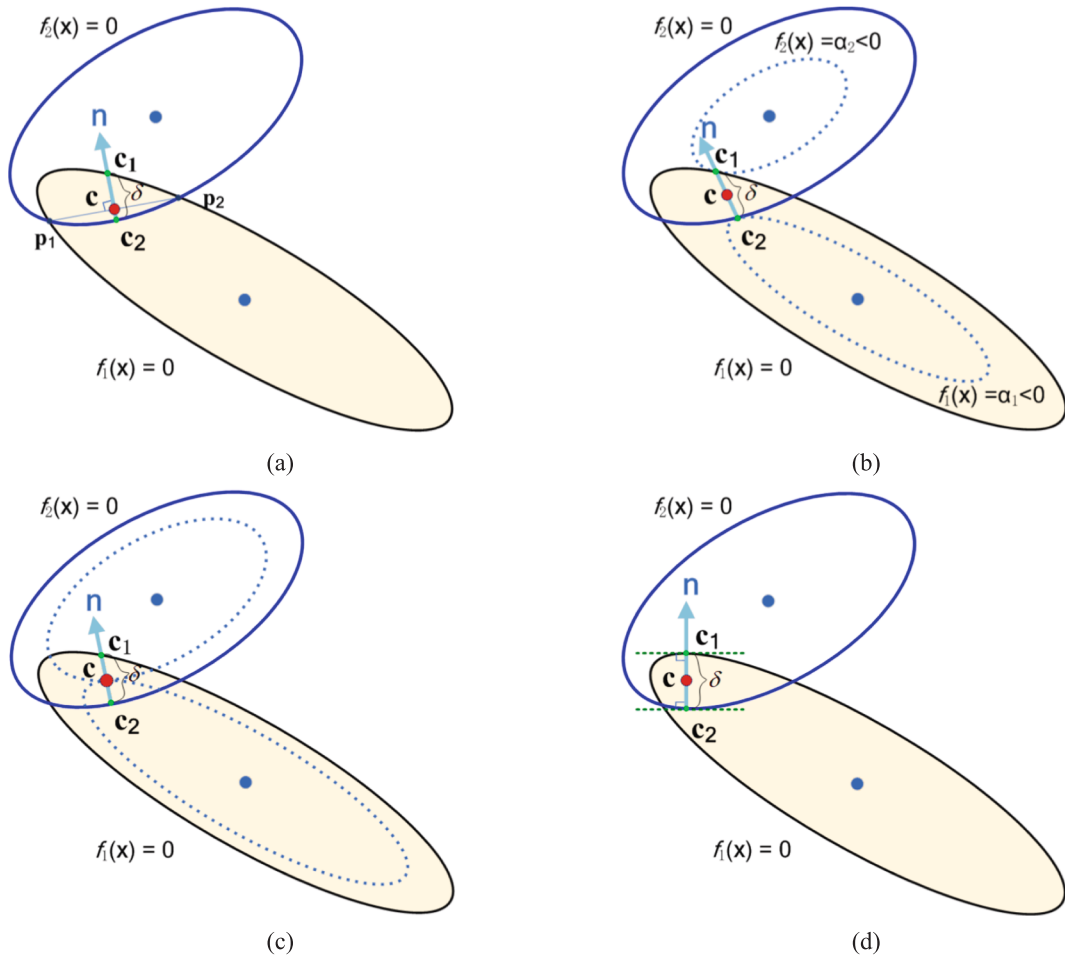


Fig. 2. Contact modelling of ellipses: (a) intersection method; (b) geometric potential method; (c) midway method; (d) common-normal method.

Marquardt (LM) method (Lourakis, 2005) for contact detection of superellipsoids. A key advantage of LM is its focus on searching for contact normal rather than two contact points, as there exists a bijective relationship between contact points on the surface and contact normal for smooth convex particles. Additionally, Zhao and Zhao (2019, 2021) proposed a more robust and efficient approach by integrating LM with erosion-Gilbert-Johnson-Keerthi (GJK) for poly-superellipsoids. However, Wachs et al. (2012) highlighted that erosion-GJK can only obtain

approximations of the true contact features, which are dependent on the shrunk scale. Lai et al. (2021) demonstrated that this can lead to an overestimation of contact overlap, though its impact on contact normal and contact point accuracy is minimal. On the contrary, the expanding polytope algorithm (EPA) (van de Bergen, 2001) provides accurate results for contact features, although some researchers note that EPA may be less efficient than erosion-GJK. Consequently, a hybrid GJK and EPA approach has been successfully applied to polygons/polyhedra (Feng

and Tan, 2020, 2021; Descantes et al., 2019; Seelen et al., 2018), which is robust and effective. Recently, this approach has been extended to smooth shapes, such as polybézier-based particles (Lai and Huang, 2021) and Fourier series-based particles (Huang et al., 2022).

Machine learning is increasingly being used in computational granular media (Qu et al., 2025). Particularly, the machine-learning-enabled DEM studies (Hwang et al., 2022; Lai et al., 2022; Huang et al., 2024) can greatly enhance the computational efficiency for contact detection and resolution of 2D/3D irregular-shaped particles.

Despite existing algorithms, an accurate, stable and efficient contact detection framework for ellipses in discrete element modelling is still lacking. The second objective of this paper is to propose such a framework. Recognizing the nice features of GJK-EPA, we further couple it with LM in this work and demonstrate its advantages over other solution strategies through some examples in Section 5. The remainder of this paper is organized as follows: Section 2 presents a detailed comparison of four widely used contact modelling methods for ellipses, and the energy-conserving property of each one is critically evaluated. Then, Section 3 introduces the contact detection and two-step contact resolution procedure of two ellipses. Additionally, a simple procedure for contact detection and resolution between ellipses and convex/non-convex boundaries is given. In Section 4, two examples with elastic impact are conducted to verify the energy-conserving property of the proposed framework for various contact scenarios (including ellipse-ellipse and ellipse-boundary). To demonstrate robustness and applicability, an additional example involving a large number of ellipses is also provided. Section 5 discusses the computational performance and advantages of the proposed framework. Finally, conclusions are drawn in Section 6.

## 2. Critical assessment of the energy-conserving property of existing contacting modelling approaches

### 2.1. Geometric description of ellipses

An ellipse is fully described by its two geometric features, semi-major axis  $a$  and semi-minor axis  $b$ , and two spatial position properties, central coordinates  $\mathbf{c}_0$  and rotation angle  $\theta$ , which is denoted as  $\xi$ :  $\{a, b, \mathbf{c}_0, \theta\}$ , see Fig. 1(a). Meanwhile, two contacting ellipses  $\xi_i$ :  $\{a_i, b_i, \mathbf{c}_i^j, \theta_i\}$  and  $\xi_j$ :  $\{a_j, b_j, \mathbf{c}_j^i, \theta_j\}$  are shown in Fig. 1(b). The ellipse can be defined as the set of roots  $\mathbf{x} = [x, y]^T$  of the geometric potential function as follows:

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{c}_0)^T \mathbf{Q} (\mathbf{x} - \mathbf{c}_0) - 1 \quad (1)$$

$$\mathbf{Q} = \mathbf{R} \mathbf{D} \mathbf{R}^T, \quad \mathbf{D} = \begin{bmatrix} 1/a^2 & 0 \\ 0 & 1/b^2 \end{bmatrix}, \quad \mathbf{R} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \theta \in [0, 2\pi] \quad (2)$$

where  $\mathbf{Q}$  is a symmetric positive-definite matrix;  $\mathbf{D}$  is a diagonal matrix with strictly positive entries;  $\mathbf{R}$  is a rotation matrix;  $\theta$  is the rotation angle of a particle from the  $x$ -axis of the global coordinate system to the  $x'$ -axis of the local coordinate system with anti-clockwise direction.

### 2.2. A brief overview of four contact modelling methods

#### 2.2.1. Intersection method (IS)

In IS, two intersection points ( $\mathbf{p}_1$  and  $\mathbf{p}_2$  in Fig. 2(a)) should be first found by solving the quartic equation resulting from combining the simultaneous equations of two ellipses as follows:

$$f_1(\mathbf{x}) = 0, \quad f_2(\mathbf{x}) = 0 \quad (3)$$

The Lin-Bairstow method can be used to solve Eq. (3) to find two intersection points. Then the contact features (i.e. contact normal  $\mathbf{n}$ , contact point  $\mathbf{c}$ , and contact overlap  $\delta$ ) are determined as follows:

- Contact point  $\mathbf{c}$ :  $\mathbf{c} = (\mathbf{p}_1 + \mathbf{p}_2)/2$ , which is the midpoint of the line joining the intersections  $\mathbf{p}_1$  and  $\mathbf{p}_2$ .
- Contact normal  $\mathbf{n}$ :  $\mathbf{n}$  is perpendicular to the segment  $\mathbf{p}_1\mathbf{p}_2$ , and  $(\mathbf{c} - \mathbf{c}_0) \cdot \mathbf{n}_1(\mathbf{c}) = (\mathbf{c} - \mathbf{c}_0) \cdot \frac{\nabla f_1(\mathbf{c})}{\|\nabla f_1(\mathbf{c})\|} = (\mathbf{c} - \mathbf{c}_0) \cdot \frac{\mathbf{Q}(\mathbf{c} - \mathbf{c}_0)}{\|\mathbf{Q}(\mathbf{c} - \mathbf{c}_0)\|} > 0$ , where  $\mathbf{n}_1(\mathbf{c})$  is the outward unit normal vector at  $\mathbf{c}$  on ellipse 1.
- Contact overlap  $\delta$ : the contact normal line will intersect the ellipse  $f_1(\mathbf{x}) = 0$  and  $f_2(\mathbf{x}) = 0$  at two points  $\mathbf{c}_1$  and  $\mathbf{c}_2$  respectively:

$$f_1(\mathbf{c}_1(\lambda_1)) = 0, \quad f_2(\mathbf{c}_2(\lambda_2)) = 0 \quad (4)$$

$$\mathbf{c}_1(\lambda_1) = \mathbf{c} + \lambda_1 \mathbf{n}, \quad \mathbf{c}_2(\lambda_2) = \mathbf{c} - \lambda_2 \mathbf{n} \quad (5)$$

where  $\lambda_1$  and  $\lambda_2$  are two non-negative parameters to be determined.

Then the contact overlap  $\delta$  is the distance between  $\mathbf{c}_1$  and  $\mathbf{c}_2$ :  $\delta = \|\mathbf{c}_1 - \mathbf{c}_2\|$ .

#### 2.2.2. Geometric potential method (GP)

As shown in Fig. 2(b), the two contact points ( $\mathbf{c}_1$  and  $\mathbf{c}_2$ ) in GP are determined in the following way:

$$\begin{cases} \mathbf{c}_1 = \underset{\forall \mathbf{x}: f_1(\mathbf{x})=0}{\operatorname{argmin}} f_2(\mathbf{x}) \\ \mathbf{c}_2 = \underset{\forall \mathbf{x}: f_2(\mathbf{x})=0}{\operatorname{argmin}} f_1(\mathbf{x}) \end{cases} \quad (6)$$

After  $\mathbf{c}_1$  and  $\mathbf{c}_2$  are obtained, the contact point  $\mathbf{c}$  is the midpoint of  $\mathbf{c}_1$  and  $\mathbf{c}_2$ , given by  $\mathbf{c} = (\mathbf{c}_1 + \mathbf{c}_2)/2$ . The contact normal  $\mathbf{n}$  is the normalized vector from  $\mathbf{c}_2$  to  $\mathbf{c}_1$ :  $\mathbf{n} = (\mathbf{c}_1 - \mathbf{c}_2)/\|\mathbf{c}_1 - \mathbf{c}_2\|$ . The contact overlap  $\delta$  is given by:  $\delta = \|\mathbf{c}_1 - \mathbf{c}_2\|$ .

To find  $\mathbf{c}_1$  and  $\mathbf{c}_2$ , Eq. (6) can be re-written as:

$$\begin{cases} \nabla f_1(\mathbf{c}_1) + \mu^2 \nabla f_2(\mathbf{c}_1) = 0 \\ f_1(\mathbf{c}_1) = 0 \end{cases} \quad (7)$$

$$\begin{cases} \nabla f_1(\mathbf{c}_2) + \nu^2 \nabla f_2(\mathbf{c}_2) = 0 \\ f_2(\mathbf{c}_2) = 0 \end{cases} \quad (8)$$

where  $\mu$  and  $\nu$  are two unknown scalars.

#### 2.2.3. Midway method (MW)

As shown in Fig. 2(c), the contact point  $\mathbf{c}$  in MW is first determined as the point that has the equal potential of  $f_1(\mathbf{x})$  and  $f_2(\mathbf{x})$  but the minimal total potential  $\operatorname{off}_1(\mathbf{x}) + f_2(\mathbf{x})$ . This can be formulated as solving a constrained optimization problem:

$$\begin{cases} \min f_1(\mathbf{x}) + f_2(\mathbf{x}) \\ \text{s.t. } f_1(\mathbf{x}) - f_2(\mathbf{x}) = 0 \end{cases} \quad (9)$$

This problem can be further re-written as follows:

$$\begin{cases} \nabla f_1(\mathbf{c}) + \mu^2 \nabla f_2(\mathbf{c}) = 0 \\ f_1(\mathbf{c}) - f_2(\mathbf{c}) = 0 \end{cases} \quad (10)$$

After determining the contact point  $\mathbf{c}$ , the contact normal  $\mathbf{n}$  is defined as the gradient of a potential function  $f_1(\mathbf{x})$  at  $\mathbf{c}$ , that is  $\mathbf{n} = \nabla f_1(\mathbf{c})/\|\nabla f_1(\mathbf{c})\|$ . The contact normal line will intersect the ellipse  $f_1(\mathbf{x}) = 0$  and  $f_2(\mathbf{x}) = 0$  at two points  $\mathbf{c}_1$  and  $\mathbf{c}_2$ , then the contact overlap  $\delta$  is  $\delta = \|\mathbf{c}_1 - \mathbf{c}_2\|$ .

#### 2.2.4. Common-normal method (CN)

As shown in Fig. 2(d), the normal vectors at the two special points ( $\mathbf{c}_1$  and  $\mathbf{c}_2$ ) in CN are parallel, but not necessarily in GP. Kildashti et al. (2018) identified limitations in the original equations of Lin and Ng (1995), where relatively larger errors may exist in the  $z$ -component.

Mathematically,  $\mathbf{c}_1$  and  $\mathbf{c}_2$  are the solutions to the following equations:

$$\begin{cases} \nabla f_1(\mathbf{c}_1) + \mu^2 (\mathbf{c}_2 - \mathbf{c}_1) = 0 \\ \nabla f_2(\mathbf{c}_2) + \nu^2 (\mathbf{c}_1 - \mathbf{c}_2) = 0 \\ f_1(\mathbf{c}_1) = 0 \\ f_2(\mathbf{c}_2) = 0 \end{cases} \quad (11)$$

**Table 1**

No. of variables within different contact detection approaches.

| Algorithm | Solution Equation | 2D               |                                                          | 3D               |                                                                |
|-----------|-------------------|------------------|----------------------------------------------------------|------------------|----------------------------------------------------------------|
|           |                   | No. of variables | Unknown variables                                        | No. of variables | Unknown variables                                              |
| IS        | Eq. (3)           | 4                | $\mathbf{p}_1(x, y)$ ,<br>$\mathbf{p}_2(x, y)$           | 6                | $\mathbf{p}_1(x, y, z)$ ,<br>$\mathbf{p}_2(x, y, z)$           |
| GP        | Eqs. (7),<br>(8)  | 6                | $\mathbf{c}_1(x, y)$ ,<br>$\mathbf{c}_2(x, y), \mu, \nu$ | 8                | $\mathbf{c}_1(x, y, z)$ ,<br>$\mathbf{c}_2(x, y, z), \mu, \nu$ |
| MW        | Eq. (10)          | 3                | $\mathbf{c}(x, y), \mu$                                  | 4                | $\mathbf{c}(x, y, z), \mu$                                     |
| CN        | Eq. (11)          | 6                | $\mathbf{c}_1(x, y)$ ,<br>$\mathbf{c}_2(x, y), \mu, \nu$ | 8                | $\mathbf{c}_1(x, y, z)$ ,<br>$\mathbf{c}_2(x, y, z), \mu, \nu$ |

After the two contact points  $\mathbf{c}_1$  and  $\mathbf{c}_2$  are obtained, the contact features can be determined:  $\mathbf{c}=(\mathbf{c}_1 + \mathbf{c}_2)/2$ ,  $\mathbf{n} = \nabla f_1(\mathbf{c}_1)/\|\nabla f_1(\mathbf{c}_1)\| = -\nabla f_2(\mathbf{c}_2)/\|\nabla f_2(\mathbf{c}_2)\|$ , and  $\delta = \|\mathbf{c}_1 - \mathbf{c}_2\|$ .

Notably, Eq. (11) yields two pairs of contact points ( $\mathbf{c}_1$  and  $\mathbf{c}_2$ ), with one pair corresponding to minimum overlap and the other to maximum overlap. Therefore, solving this equation demands meticulous care to avoid obtaining incorrect solutions.

### 2.3. Comparison of contact features across four contact modelling methods

Table 1 compares the number of variables required by different contact detection approaches. Among the four approaches, MW has the fewest unknown variables, whereas GP and CN require the most. Consequently, MW is the most computationally effective approach and has been widely adopted. Overall, the efficiency order of the four approaches is as follows: MW > IS > GP > CN.

Fig. 3 illustrates the differences in the location of contact points between two contact ellipses  $\xi_1$ : {10, 4.1, (0, 0), -0.5} and  $\xi_2$ : {10, 4.1, (4, 5), -1.0}. The corresponding contact points and contact features using four approaches are given in Table 2.

Furthermore, Fig. 4 illustrates differences in the location of contact points between ellipse  $\xi_1$ : {10, 4.1, (0, 0), -0.5} and line  $W_1$ : {(-10,

-4), (10, -4)}. The corresponding contact points and contact features using four approaches are given in Table 3. From the above two examples, it is evident that there is a relatively large difference among these approaches.

### 2.4. Energy-conserving property of these four methods

As shown in Section 2.3, these four contact modelling approaches give significantly different contact geometry features. Are any of these methods theoretically correct with energy-conserving? To address this question, the following deduction is provided.

Following the theoretical framework developed by Feng and Tan (2021), an overlap-based contact model with an energy-conserving property requires the following relationship to hold

$$\frac{\partial \delta}{\partial \mathbf{x}} = -\mathbf{n} \quad (12)$$

where  $\mathbf{x}$  is taken to be the coordinates of a reference point on convex shape  $B$ .

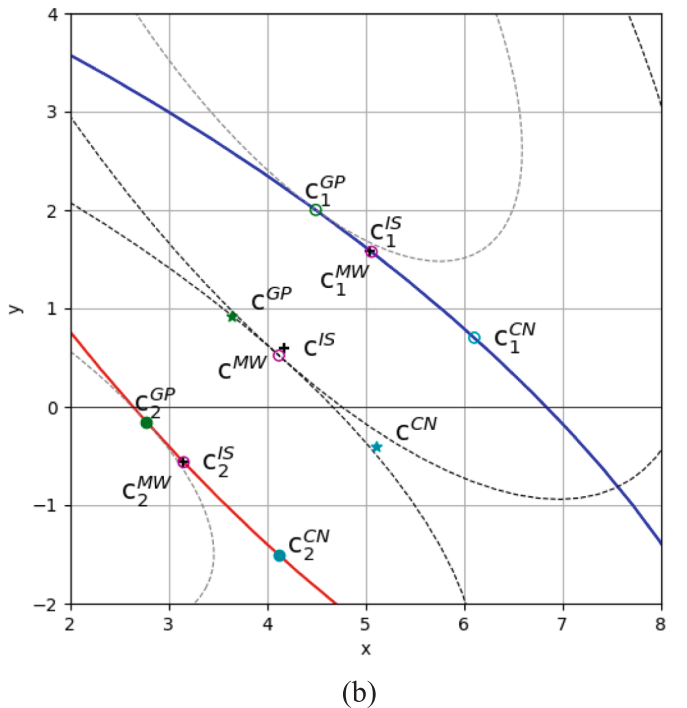
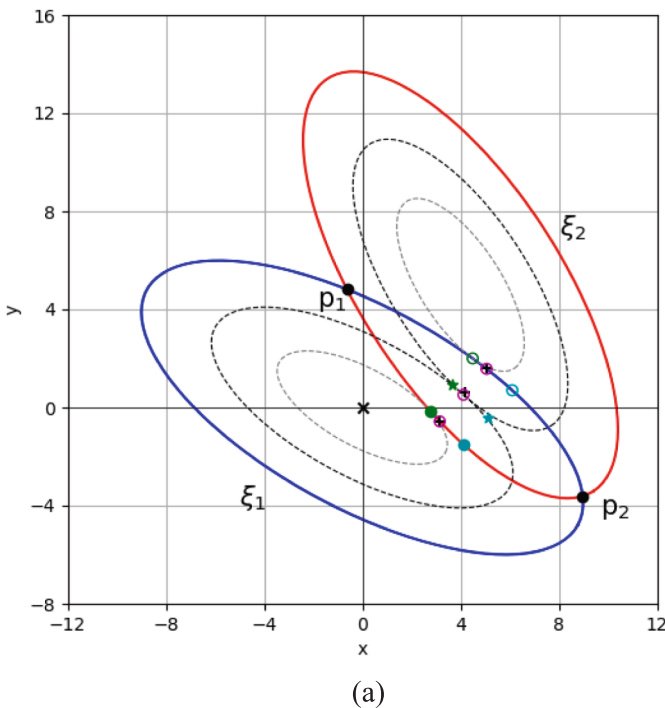
The clear geometric explanation of Eq. (12) is that the most effective

**Table 2**

Comparison of contact features between two ellipses using four contact detection approaches.

| Algorithm | Contact point $\mathbf{c}_1$ at ellipse $\xi_1$ | Contact point $\mathbf{c}_2$ at ellipse $\xi_2$ | Contact features           |                             |                          |
|-----------|-------------------------------------------------|-------------------------------------------------|----------------------------|-----------------------------|--------------------------|
|           |                                                 |                                                 | Contact point $\mathbf{c}$ | Contact normal $\mathbf{n}$ | Contact overlap $\delta$ |
| IS        | (5.0463, 1.5897)                                | (3.1557, -0.5629)                               | (4.1790, 0.6022)           | (0.6598, 0.7513)            | 2.8650                   |
| GP        | (4.4973, 2.0017)                                | (2.7785, -0.1543)                               | (3.6379, 0.9236)           | (0.6233, 0.7819)            | 2.7573                   |
| MW        | (5.0639, 1.5760)                                | (3.1556, -0.5628)                               | (4.1233, 0.5217)           | (0.6657, 0.7461)            | 2.8664                   |
| CN        | (6.1083, 0.7025)                                | (4.1185, -1.5037)                               | (5.1134, -0.4005)          | (0.6697, 0.7425)            | 2.9710                   |

Note: Two intersection points in IS are respectively  $\mathbf{p}_1$  (-0.6189, 4.8159) and  $\mathbf{p}_2$  (8.9770, -3.6115).



**Fig. 3.** Location of the contact points between two ellipses using different contact detection approaches: (a) whole view; (b) zoom-in view.



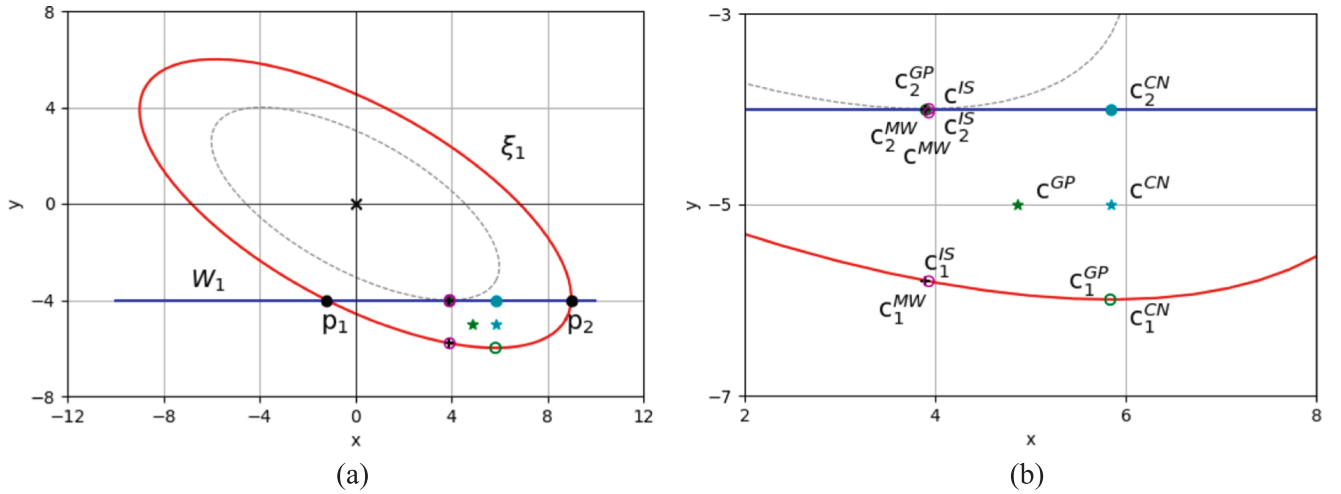


Fig. 4. Location of the contact points between ellipse and line using different contact detection approaches: (a) whole view; (b) zoom-in view.

Table 3

Comparison of contact features between ellipse and line using four contact detection approaches.

| Algorithm | Contact point $c_1$ on ellipse $\xi_1$ | Contact point $c_2$ on line $W_1$ | Contact features  |                    |                          |
|-----------|----------------------------------------|-----------------------------------|-------------------|--------------------|--------------------------|
|           |                                        |                                   | Contact point $c$ | Contact normal $n$ | Contact overlap $\delta$ |
| IS        | (3.8964, -5.7950)                      | (3.8964, -4.0)                    | (3.8964, -4.0)    | (0.0, 1.0)         | 1.7950                   |
| GP        | (5.8390, -5.9942)                      | (3.8964, -4.0)                    | (4.8677, -4.9971) | (-0.6977, 0.7163)  | 2.7840                   |
| MW        | (3.9344, -5.8021)                      | (3.9344, -4.0)                    | (3.9344, -4.0389) | (0.0, 1.0)         | 1.8021                   |
| CN        | (5.8390, -5.9942)                      | (5.8390, -4.0)                    | (5.8390, -4.9971) | (0.0, 1.0)         | 1.9942                   |

Note: Two intersection points in IS are respectively  $p_1$  (-1.1977, -4.0) and  $p_2$  (8.9906, -4.0).

way to move  $B$  to reduce the overlap  $\delta$  must be along the negative direction of the contact normal  $n$ . It should be highlighted that Eq. (12) is held for both convex and concave shapes.

From the Minkowski difference of two convex shapes  $A$  and  $B$  (i.e.  $A \ominus B$ ) in contact, the contact overlap  $\delta$  can be defined as the shortest distance from the origin  $o$  to the boundary of  $A \ominus B$  (i.e.  $\partial(A \ominus B)$ ):

$$\delta = \min_{c \in \partial(A \ominus B)} \|c - o\| = \|c_M\| \quad (13)$$

where  $c$  is a point on  $\partial(A \ominus B)$ ;  $c_M$  is the orthogonal projection of  $o$  onto  $\partial(A \ominus B)$ , from which the normal direction can be attained  $n = c_M/\delta$ .

The two corresponding contact points associated with  $c_M$  on  $\partial A$  and  $\partial B$  are respectively  $c_1$  and  $c_2$ , from which the contact point  $c$  can be chosen as the midpoint:  $c = (c_1 + c_2)/2$ .

Obviously,  $n$  is normal of  $\partial(A \ominus B)$  at  $c_M$ . Both  $n$  and the line  $c_1c_2$  are the common normal to both  $A$  and  $B$ . For these four methods, this essentially demonstrates that CN is theoretically correct and thus energy-conserving.

To further illustrate the differences, the corresponding Minkowski difference of  $W_1$  and  $\xi_1$  in Fig. 4(a) is shown in Fig. 5(a). In this contact scenario,  $\delta = 1.9942$ ,  $n = (0.0, 1.0)$ , and  $c_M = (0, 1.9942)$ . The corresponding  $c_M$  for the four methods is displayed in Fig. 5(b), which are  $c_M^{IS} = (0, 1.7950)$ ,  $c_M^{GP} = (-1.9426, 1.9942)$ ,  $c_M^{MW} = (0, 1.8021)$ ,  $c_M^{CN} = (0, 1.9942)$ . It is evident that only CN can get the correct result. The contact overlap associated with MW and IS is less than the correct value, while the contact overlap associated with GP is larger than the correct value. As explained previously, only the CN method is theoretically sound and energy-conserving, providing correct contact overlap.

In this section, we demonstrated that only the CN method is energy-conserving, while the other three methods may not. This occurs because

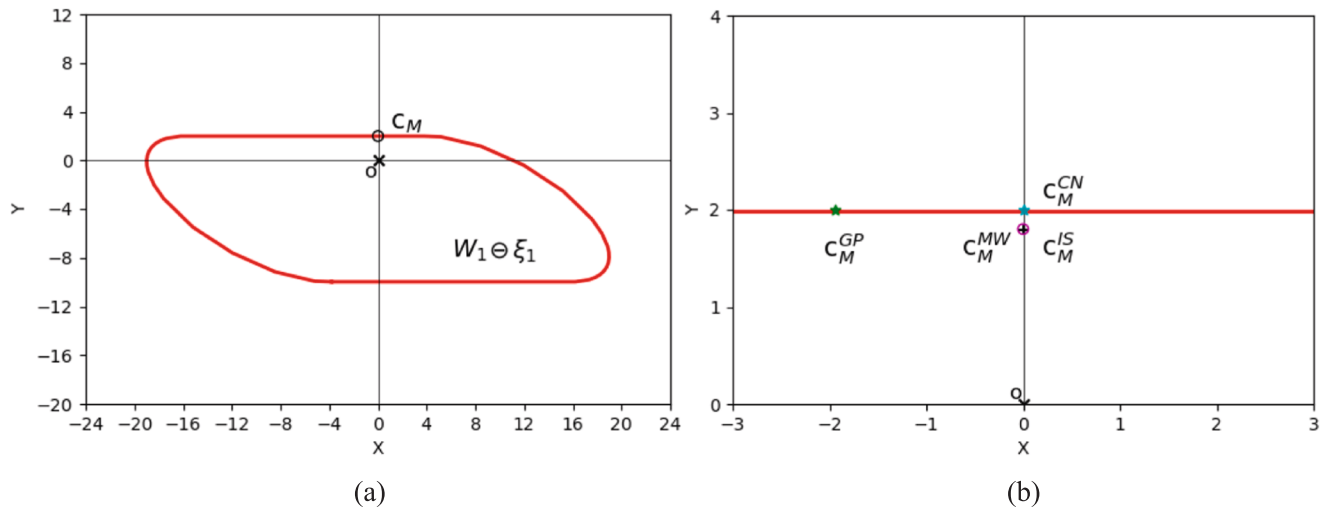


Fig. 5. The corresponding Minkowski difference of  $W_1$  and  $\xi_1$ : (a) whole view; (b) zoom-in view.

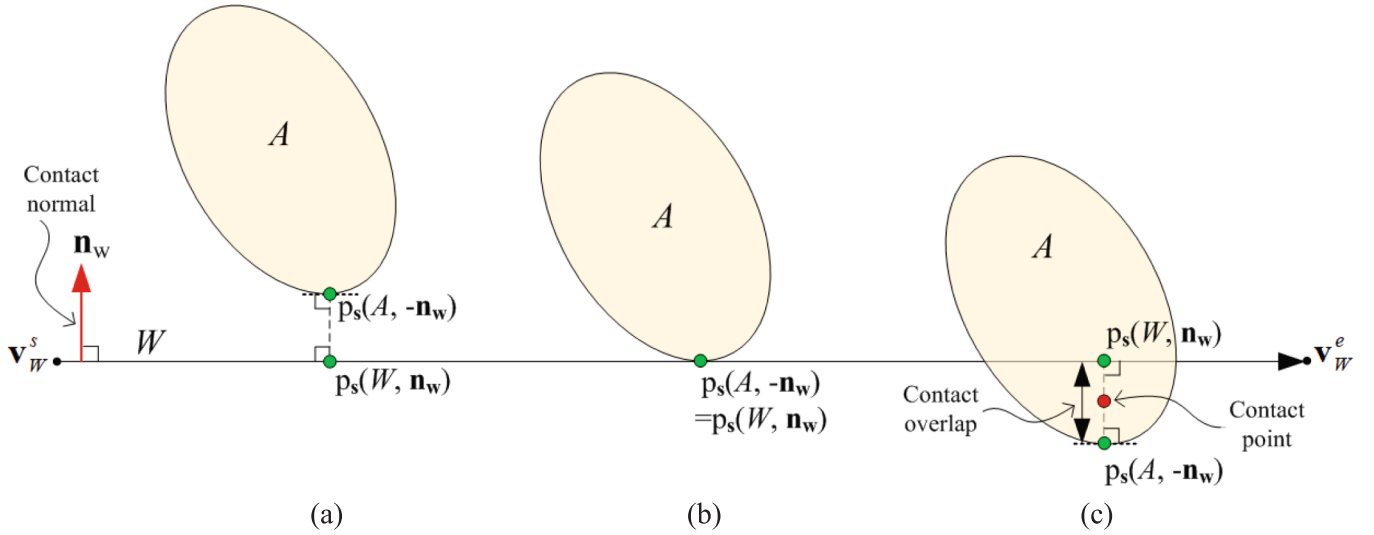


Fig. 6. Contact detection between ellipse  $A$  and line  $W$ : (a) separation; (b) touch; (c) overlap.

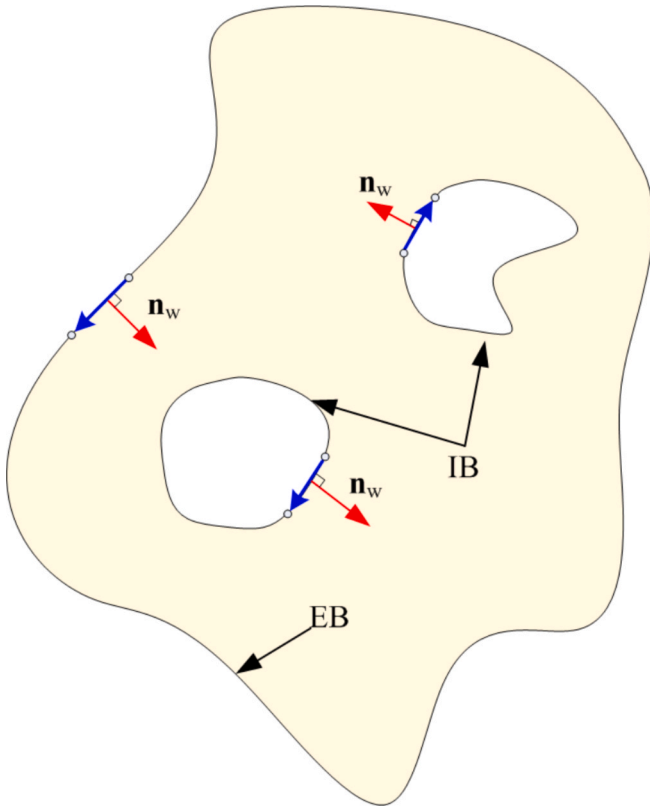


Fig. 7. The boundaries of a planar domain with two interior cavities.

that the contact normal  $\mathbf{n}$  (i.e. line  $\mathbf{c}_1\mathbf{c}_2$ ) is not always perpendicular to two ellipses at their contact points ( $\mathbf{c}_1$  and  $\mathbf{c}_2$ ). As a consequence, the other three methods cannot consistently guarantee that the contact overlap satisfies the Eq. (12). For this reason, the CN method is used for contact detection and resolution for ellipses in this work. However, since particles in DEM computations undergo minimal deformation, the contact overlap is typically very small. Consequently, the final contact features obtained from these four different approaches may exhibit only insignificant differences. Thus, these alternative approaches can be still

adopted in DEM.

### 3. Contact detection and resolution of two ellipses

#### 3.1. Contact detection

As stated in Feng and Tan (2020, 2021), the contact state between the two convex blocks can be determined by checking if the origin is enclosed within their Minkowski difference. The explicit constructing of the Minkowski difference of two convex blocks can refer to Xia et al. (2023). While obtaining the contact state through this method is computationally intensive, GJK originally proposed by Gilbert et al. (1988) offers a more elegant and efficient alternative by only checking if the origin is enclosed in the Minkowski difference through iterations using support points of the convex blocks. The specifics of GJK are outlined in Algorithm 1, where the support point of ellipse  $A$  along direction  $\mathbf{n}=(n_x, n_y)$  can be explicitly defined:

$$\mathbf{p}_s(A, \mathbf{n}) = \mathbf{R}\mathbf{p}_s^{\text{local}}(A, \mathbf{R}^{-1}\mathbf{n}) + \mathbf{c}_0 \quad (14)$$

$$\mathbf{p}_s^{\text{local}}(A, \mathbf{n}) = \frac{(a^2 n_x, b^2 n_y)}{\|(a n_x, b n_y)\|} \quad (15)$$

where  $\mathbf{R}^{-1}$  is the inverse matrix of the rotation matrix  $\mathbf{R}$  defined in Eq. (2), that is satisfying  $\mathbf{R}^{-1} = \mathbf{R}^T$ .

Furthermore, prior to any overlap checking using GJK, a broader contact detection can be used to exclude those non-contacting ellipses, thereby enhancing the efficiency of the overlap check (Xia et al., 2023). For each ellipse, a bounding circle with a radius equal to its semi-major axis is assigned. Then the central distance  $d_{ij}$  between ellipse  $i$  with semi-major axis  $a_i$  and ellipse  $j$  with semi-major axis  $a_j$  is calculated. Three cases arise:

- (1) If  $d_{ij} \geq a_i + a_j$ , two ellipses do not overlap, and no further checks are needed.
- (2) If  $d_{ij} < b_i + b_j$ , contact exists between them.
- (3) If  $b_i + b_j \leq d_{ij} < a_i + a_j$ , further checks are required to determine the specific contact state.

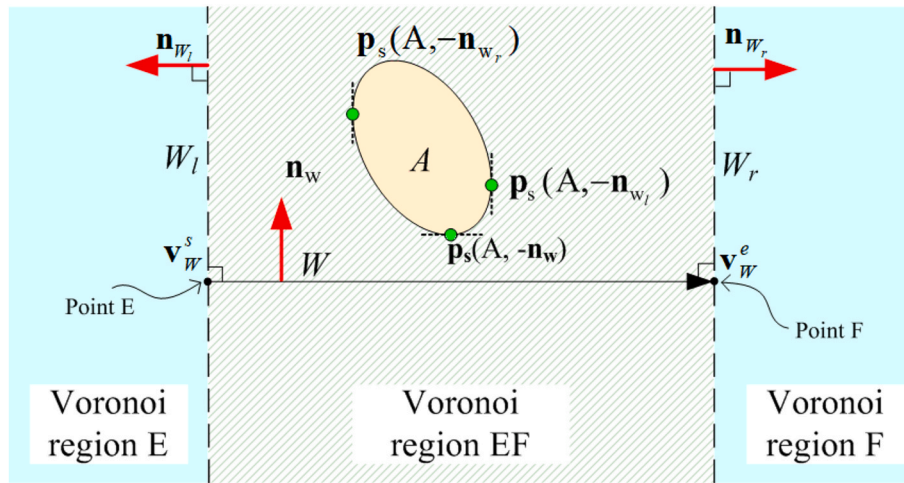


Fig. 8. A segment EF and its Voronoi regions.

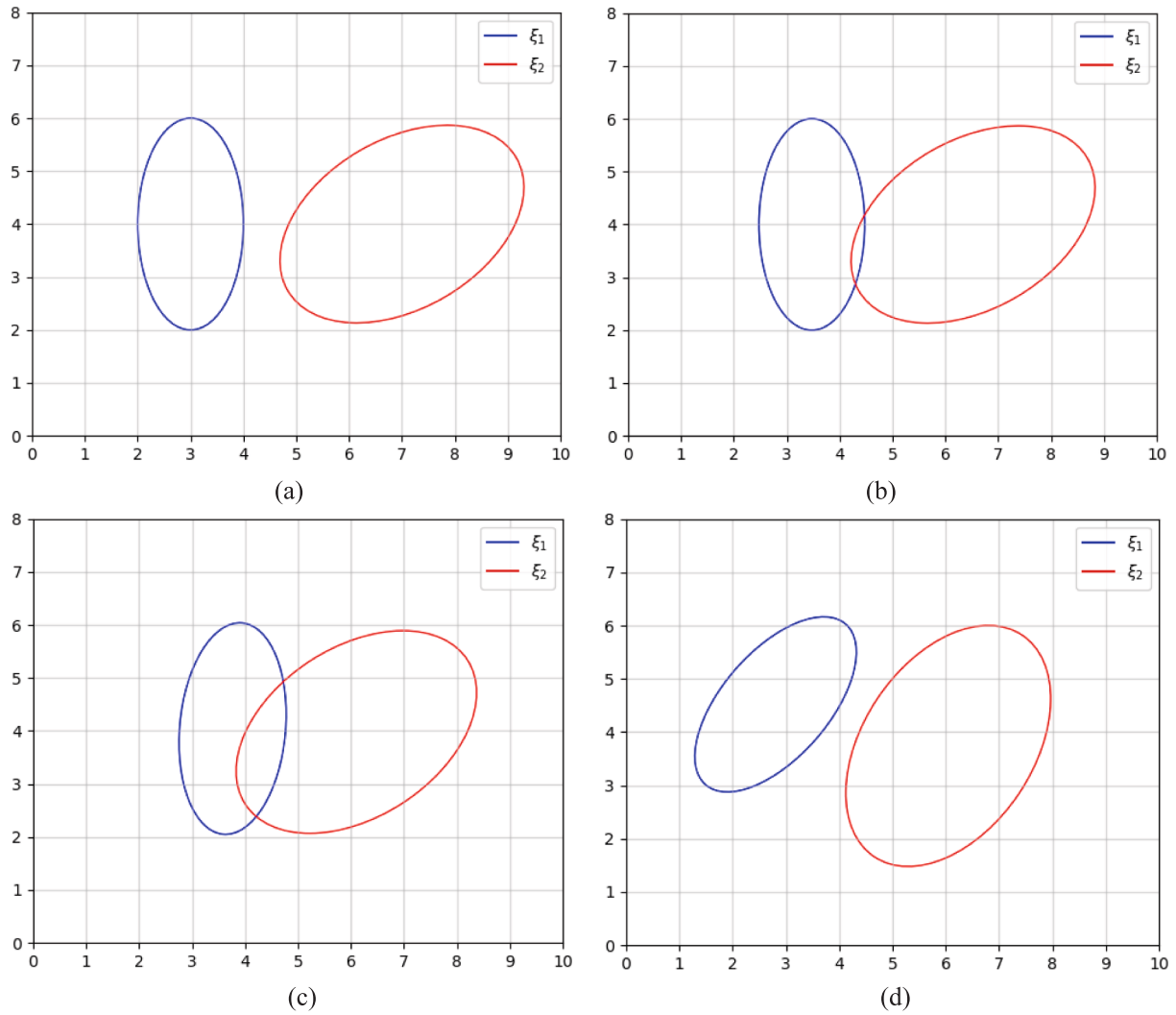


Fig. 9. Elastic impact of two ellipses: (a) before impact; (b) at the increasing penetration stage; (c) at the maximum penetration; (d) after impact.

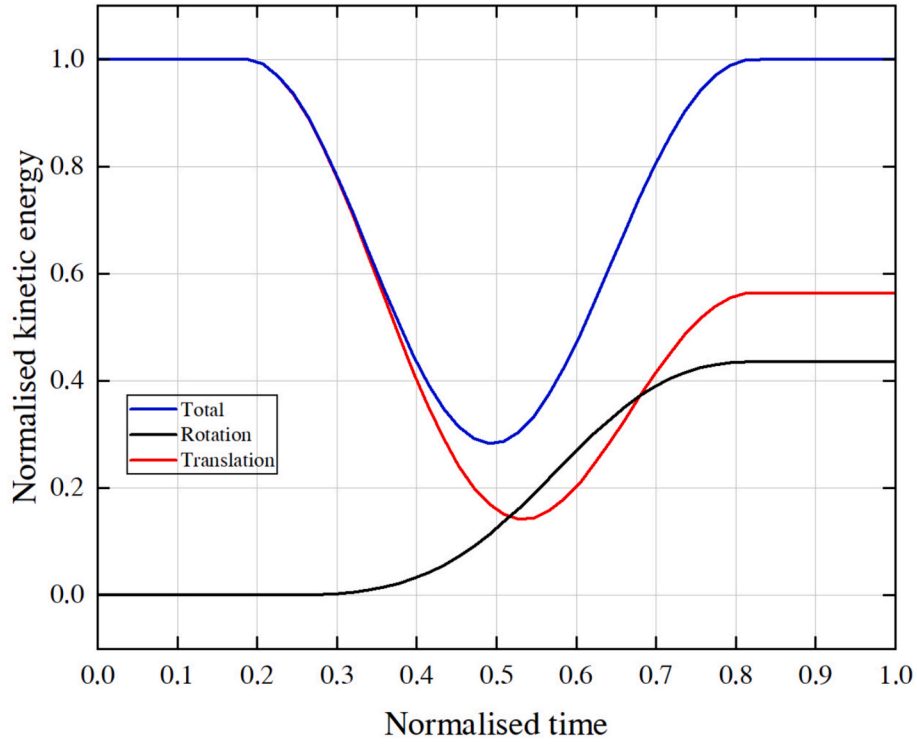


Fig. 10. Elastic impact of two ellipses: evolution of translational, rotational and total kinetic energy.

#### Algorithm 1. GJK

**Input:** Two ellipses  $A$  and  $B$

**Output:** Contact state between  $A$  and  $B$

1: Initialization

a. Set Simplex  $S$  to be empty:  $S = \emptyset$

b. Choose an (arbitrary) initial search direction  $\mathbf{v} \neq \mathbf{0}$  (It is suggested to be the joint vector of the centroids of objects  $A$  and  $B$ )

2: Iteration

a. Compute the support point  $\mathbf{a} = \mathbf{p}_s(A \ominus B, \mathbf{v})$

b. If  $\mathbf{a} \cdot \mathbf{v} < 0$ ,  $S = \emptyset$ , **output (NO CONTACT)**, **stop**

c. Update Simplex  $S$

i. Add  $\mathbf{a}$  to  $S$  as a new vertex

ii. If  $S$  contains the origin  $\mathbf{o}$ , **output (CONTACT)**, **stop**

iii. Reduce  $S$  to the lowest dimension possible that still contains the closest feature to the origin by discarding vertices

iv. Compute a new search direction  $\mathbf{v}$  that is toward the origin normal to the new simplex; and GOTO 2.a

obtained, all other contact features (e.g., contact overlap and contact point) are uniquely determined. Unlike traditional methods for calculating two contact points, LM only needs to search for one variable.

By introducing a local coordinate system, the contact normal  $\mathbf{n}$  can be defined by a single parameter (i.e. contact angle  $\alpha$ ) as

$$\mathbf{n}(\alpha) = \cos\alpha \mathbf{i} + \sin\alpha \mathbf{j} \quad (16)$$

This formulation transforms the origin problem of Eq. (13) into the following global optimization problem:

$$\delta = \min_{\mathbf{c} \in \partial(A \ominus B)} \|\mathbf{c} - \mathbf{o}\| = \min_{\alpha} \|\mathbf{p}_s(A \ominus B), \mathbf{n}(\alpha)\| \quad (17)$$

In this framework, only one intermediate variable (i.e. contact angle  $\alpha$ ) is stored, which can significantly reduce memory requirements.

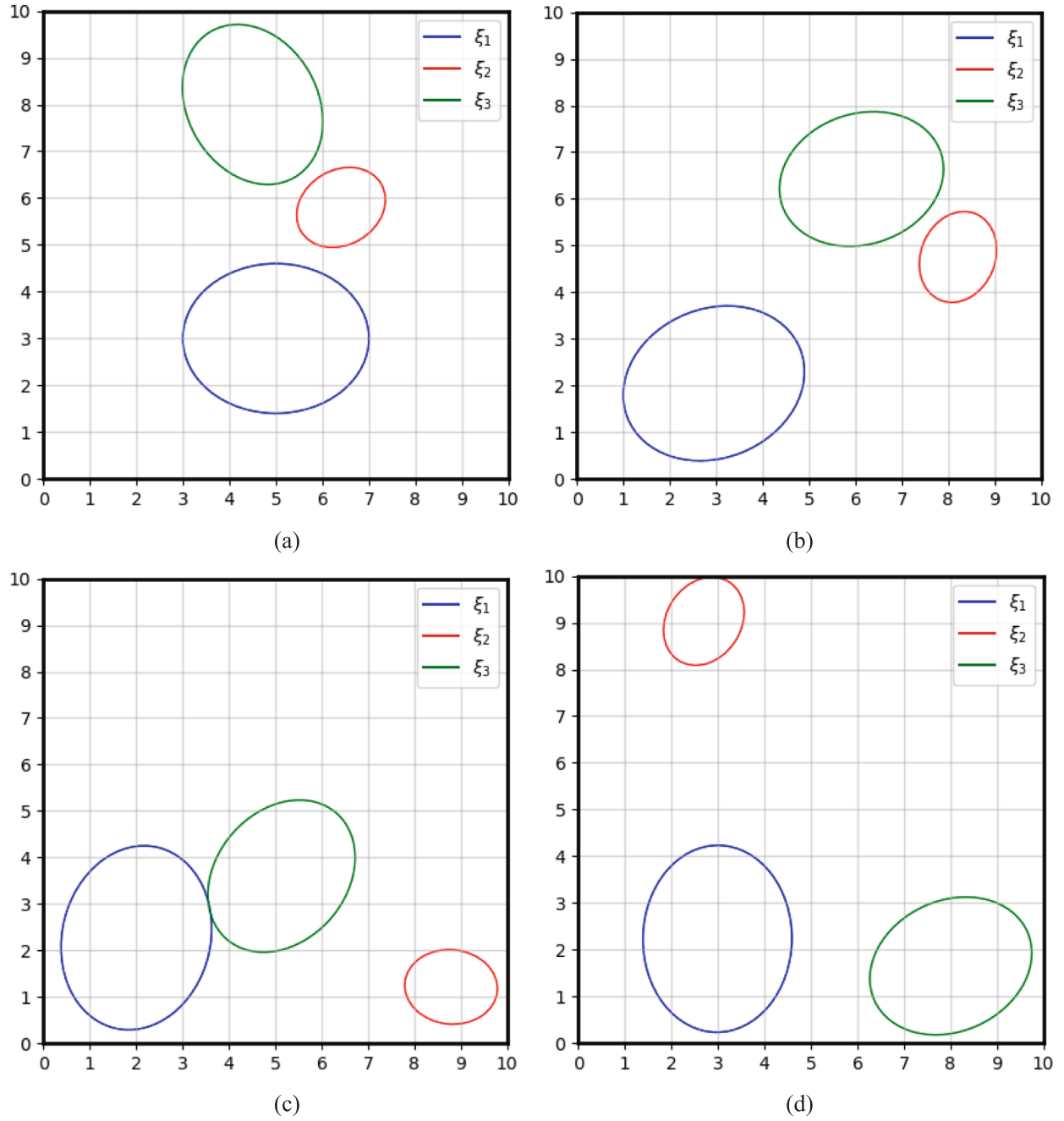
Although EPA can be employed alone to compute contact features, it may become numerically inefficient at each time step. Given that the time step in DEM simulations is normally sufficiently small to ensure conditional stability in explicit time integrations, contact pairs experience minimal spatial changes during one time step. Leveraging this temporal coherence, the contact angle from the previous step is utilized as an initial guess for the current step. In this work, contact pairs of ellipses are categorized into two types: new contacts (not in contact at the previous time step) and old contacts (in contact at the previous time step). For new contact pairs, a hybrid EPA-LM approach is employed, where EPA provides an initial guess for LM. On the contrary, for old contacts, LM takes the previous contact angle as input to compute the current contact angle and related contact features. The computational efficiency and advantages of the proposed method over other solution strategies, such as the hybrid GJK-EPA, are detailed in Section 5.

In Algorithm 3, this study initializes the user-defined parameters as  $\nu$

### 3.2. Contact resolution

After GJK identifies a contact between convex blocks, the contact features (contact overlap  $\delta$ , contact point  $\mathbf{c}$  and contact normal  $\mathbf{n}$ ) must be determined to accurately compute contact forces and moments. The proposed method involves a two-step resolution scheme:

- (1) Step 1, Use the simplex containing the origin  $\mathbf{o}$  returned from GJK as the initial polytope for EPA. EPA iteratively expands this inner polyhedral approximation of the Minkowski difference until the shortest distance converges within a specific tolerance. The detailed EPA procedure is outlined in Algorithm 2.
- (2) Step 2, Input the rough contact normal from EPA into LM (Algorithm 3) for refinement. Once the accurate contact normal is



**Fig. 11.** Elastic impact of three ellipses in a confined box: (a)  $t = 0$ ; (b)  $t = 3$  s; (c)  $t = 5$  s; (d)  $t = 8$  s.



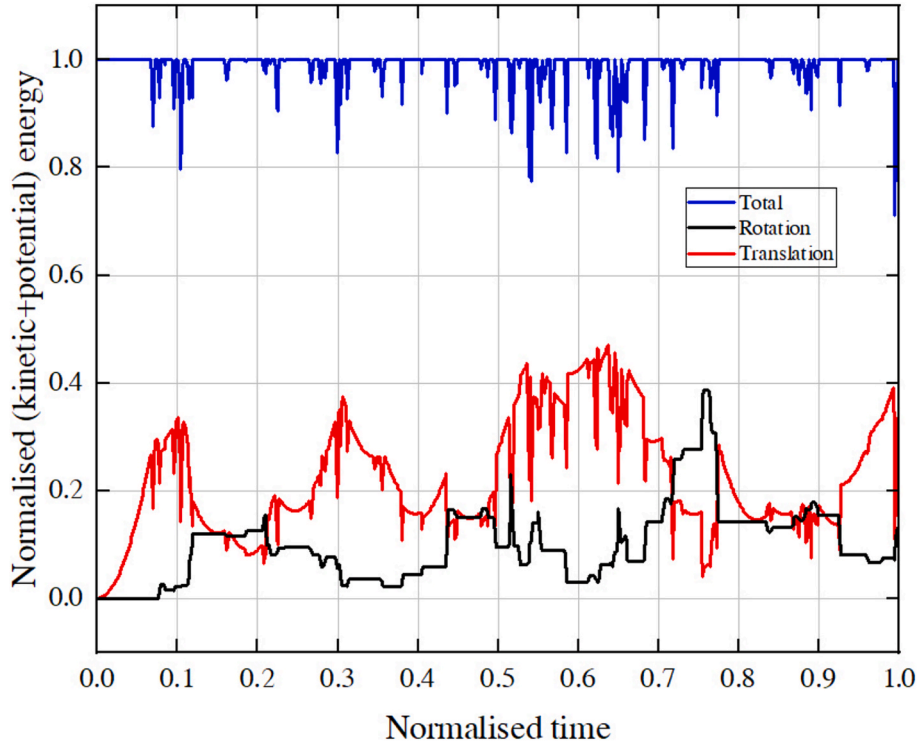


Fig. 12. Elastic impact of three ellipses in a box: evolution of translational, rotational and total energy. The total energy includes the potential energy.

$= 2.0$ ,  $\tau = 10^{-3}$ ,  $\varepsilon_1 = \varepsilon_2 = 10^{-15}$ , following [Lourakis \(2005\)](#) and [Madsen et al. \(2004\)](#)'s recommendations to achieve a stable and fast convergence.

#### Algorithm 2. EPA

**Input:** Two ellipses  $A$  and  $B$ , Simplex  $S$  outputted from GJK  
**Output:** Contact normal between  $A$  and  $B$  (i.e.  $\mathbf{n}_f$ )  
 1: Initialization  
 a. Take the simplex  $S$  returned from the GJK algorithm as the start polytope  $P=S$   
 b. Expand  $P$  to be a full simplex by adding new vertices if  $S$  is a degenerated simplex  
 c. Compute the distance  $d_i$  of each face  $i \in P$  to the origin  
 2: Iteration  
 a. Find the closest face  $f \in P$  to the origin  $\mathbf{o}$ , and set the search direction to be  $\mathbf{n}_f$ , where  $\mathbf{n}_f$  is the outward normal of the closest face  $f$ . Let  $d_f$  be the distance of the face  $f$  to the origin.  
 b. Compute the support point  $\mathbf{a} = \mathbf{p}_s(A \oplus B, \mathbf{n}_f)$   
 c. Compute the distance between  $\mathbf{a}$  and the origin  $\mathbf{o}$  along the direction  $\mathbf{n}_f$ :  $d_a = \mathbf{n}_f \cdot \mathbf{a}$   
 d. Check for convergence: if  $d_a - d_f < \varepsilon$  (given tolerance), **output** ( $\mathbf{n}_f$ ), **stop**  
 e. Expand  $P$ :  
 For each face  $f \in P$  which is visible from  $\mathbf{a}$ , that is,  $\mathbf{n}_f \cdot (\mathbf{a} - \mathbf{v}_f) > 0$  (where  $\mathbf{v}_f$  is any vertex of the face)  
 i. Remove the face  $f$  from  $P$   
 ii. Add new faces formed by the vertices of the face and the new point  $\mathbf{a}$  to  $P$ ; and calculate the distance of each new face to the origin  
 f. Go back to 2.a

#### Algorithm 3. 2D Levenberg–Marquardt algorithm

**Input:** Two ellipses  $A$  and  $B$ , contact normal  $\mathbf{n}_f$  outputted from EPA (new contact) OR contact angle  $\alpha$  from previous step (old contact)  
**Output:** Contact features (i.e.  $\delta$ ,  $\mathbf{n}$ , and  $\mathbf{c}$ ), and current contact angle  $\alpha$  between  $A$  and  $B$   
 1: Initialization  
 a. Set  $\nu$  (a factor to increasing  $\mu$ -values rapidly),  $\tau$  (a very small user-chosen value),  $\varepsilon_1$  and  $\varepsilon_2$  (two given tolerances)  
 b. Compute the initial contact angle  $\alpha$  by  $\mathbf{n}_f$  outputted from EPA (new contact) OR set from the previous step (old contact)  
 c. Compute the support point  $\mathbf{a} = \mathbf{p}_s(A \oplus B, \mathbf{n})$ ,  $\mathbf{n} = [\cos\alpha, \sin\alpha]$   
 2: Iteration  
 a. Compute the Jacobian matrix  $\mathbf{J} = \partial \mathbf{a} / \partial \alpha$ ,  $\mathbf{H} = \mathbf{J}^T \mathbf{J}$ , and  $\mathbf{G} = -\mathbf{J}^T \mathbf{a}$

(continued on next column)

(continued)

#### Algorithm 3. 2D Levenberg–Marquardt algorithm

b. Check for convergence: if  $|\mathbf{G}| < \varepsilon_1$  (given tolerance), GOTO 3  
 c. Compute the initial damping factor  $\mu = \tau \cdot \mathbf{H}$   
 d. Loop:  
 i. Compute  $\Delta\alpha$  with  $(\mathbf{H} + \mu)\Delta\alpha + \mathbf{G} = 0$   
 ii. Check for convergence: if  $|\Delta\alpha/\alpha| < \varepsilon_2$  (given tolerance), GOTO 3  
 iii. Compute contact normal  $\mathbf{n}' = [\cos(\alpha + \Delta\alpha), \sin(\alpha + \Delta\alpha)]$ , and the support point  $\mathbf{a}' = \mathbf{p}_s(A \oplus B, \mathbf{n}')$   
 iv. Compute  $\rho = (||\mathbf{a}'|| - ||\mathbf{a}'||) / (\Delta\alpha \cdot (\mu\Delta\alpha + \mathbf{G}))$   
 if  $(\rho > 0)$ : Update  $\mu = \mu \cdot \max\{1/3, 1 - (2\rho - 1)^3\}$ ,  $\nu = 2$ ,  $\alpha = \alpha + \Delta\alpha$ ,  $\mathbf{a} = \mathbf{a}'$ , GOTO 2.  
 a  
 else: Update  $\mu = \mu \cdot \nu$ ,  $\nu = 2\nu$   
 end  
 Until  $(\rho > 0)$   
 3: Compute contact features and store variable  
 a. Store the current contact angle  $\alpha$ , and compute contact normal  $\mathbf{n} = [\cos\alpha, \sin\alpha]$   
 b. Compute contact points on the two shapes:  $\mathbf{c}_1 = \mathbf{p}_s(A, \mathbf{n})$ ,  $\mathbf{c}_2 = \mathbf{p}_s(B, -\mathbf{n})$   
 c. Compute the common contact point  $\mathbf{c} = (\mathbf{c}_1 + \mathbf{c}_2)/2$ , and contact overlap  $\delta = ||\mathbf{c}_1 - \mathbf{c}_2||$

### 3.3. Special case: contact between ellipses and boundary lines

Besides ellipse-ellipse contacts, ellipses in DEM simulations can also contact with boundaries. These boundaries are often discretized into multiple lines. The proposed framework is also capable of detecting contact states and computing the contact features between ellipses and boundary lines, though this can be done in a simpler manner. For example, consider an ellipse  $A$  and a line  $W$  as shown in Fig. 6, there are three possible contact states that can occur between them: separation, touch, and overlap. Unlike ellipse-ellipse contacts, there is no need for an iterative process to find the contact normal in this case. The contact normal between a particle and a line is *priori* known, which is normal to the line. To determine the specific contact state, first, identify the support point of the ellipse in the negative contact normal, and then calculate the corresponding support point on the line as the orthogonal projection of the support point of the ellipse onto the line.

The Minkowski distance between  $W$  and  $A$ , denoted as  $\mathbf{d}(W, A)$ , is

defined as:

$$\mathbf{d}(W, A) = \mathbf{p}_s(W \ominus A, \mathbf{n}_w) = \mathbf{p}_s(W, \mathbf{n}_w) - \mathbf{p}_s(A, -\mathbf{n}_w) \quad (18)$$

where  $\mathbf{n}_w$  is the contact normal, defined as the unit vector orthogonal to line  $W$ .

The gap (when separation) or overlap (when overlap) between  $W$  and  $A$ , denoted as  $\delta(W, A)$ , is calculated as:

$$\begin{aligned} \delta(W, A) &= \mathbf{n}_w \cdot \mathbf{d}(W, A) = \mathbf{n}_w \cdot (\mathbf{p}_s(W, \mathbf{n}_w) - \mathbf{p}_s(A, -\mathbf{n}_w)) \\ &= \mathbf{n}_w \cdot (\mathbf{v}_w^s - \mathbf{p}_s(A, -\mathbf{n}_w)) \end{aligned} \quad (19)$$

where  $\mathbf{v}_w^s$  is the start point of line  $W$  (it can be any point in line  $W$ ).

Consequently, the contact state between ellipse  $A$  and line  $W$  is simplified to the following statement:

$$\begin{cases} \delta(W, A) > 0 & \equiv \text{overlap} \\ \delta(W, A) = 0 & \equiv \text{touch} \\ \delta(W, A) < 0 & \equiv \text{separation} \end{cases} \quad (20)$$

Once a contact is detected between an ellipse and a line, the corresponding contact features can be easily determined. As shown in Fig. 6 (c), after obtaining two support points (i.e.  $\mathbf{p}_s(W, \mathbf{n}_w)$  and  $\mathbf{p}_s(A, -\mathbf{n}_w)$ ), the contact point  $\mathbf{c}$  is set as their middle point:

$$\mathbf{c} = (\mathbf{p}_s(W, \mathbf{n}_w) + \mathbf{p}_s(A, -\mathbf{n}_w))/2 \quad (21)$$

Finally, all contact features, such as contact point  $\mathbf{c}$ , contact overlap  $\delta(W, A)$ , and contact normal  $\mathbf{n}$ , are then obtained.

As shown in Fig. 7, the external boundary (EB) of the contact area is discretized into many connect lines with anti-clockwise direction, such that the outer normal direction  $\mathbf{n}_w$  points toward the interior of the contact area. Conversely, if the interior boundary (IB) is considered, it must be discretized into many connect lines with a clockwise direction, ensuring that the outer normal direction  $\mathbf{n}_w$  points into the interior of the contact area.

It should be noted that if EB or IB is convex, the contact state between ellipse  $A$  and line  $W$  can be determined using Eq. (20). However, if EB/IB is concave, a concept from computational geometry (i.e. Voronoi regions) is important for designing intersection tests. As shown in Fig. 8, a segment EF has three associated Voronoi regions: two vertex regions (region E, region F), and one edge region (region EF). The line segment connects vertices E (positioned at  $\mathbf{v}_w^s$ ) and F (positioned at  $\mathbf{v}_w^e$ ).

The ellipse  $A$  is located in the Voronoi region EF associated with the line  $W$  if the following condition is satisfied:

$$\begin{cases} \mathbf{n}_{w_l} \cdot (\mathbf{v}_w^s - \mathbf{p}_s(A, -\mathbf{n}_{w_l})) \geq 0 \\ \mathbf{n}_{w_r} \cdot (\mathbf{v}_w^e - \mathbf{p}_s(A, -\mathbf{n}_{w_r})) \geq 0 \end{cases} \quad (22)$$

where  $\mathbf{v}_w^e$  is the end point of the line  $W$ ;  $\mathbf{n}_{w_l}$  is the unit vector normal to the line  $W_l$  formed by rotating  $\mathbf{n}_w$  90° anticlockwise as shown in Fig. 8;  $\mathbf{n}_{w_r}$  is the unit vector normal to the line  $W_r$  formed by rotating  $\mathbf{n}_w$  90° clockwise as shown in Fig. 8.

It is important to note that this approach is suitable not only for contact between line and ellipse, but also for contact between a line and arbitrary convex particles with an explicit support function.

#### 4. Numerical examples

In this section, numerical examples are provided to verify the proposed framework, and demonstrate the applicability and flexibility of the approach. The approach is programmed using Fortran95 in an in-house code. Computing resources used in this work and associated

corresponding computational costs are also given as follows: (a) Processor: Intel Core i7-10700 CPU @2.90 GHz; (b) Memory: 16 GB.

##### 4.1. Energy conservation verification with two ellipses

The first example involves an elastic impact of two ellipses  $\xi_1$ : {2, 1, (3, 4),  $\pi/2$ } and  $\xi_2$ : {2.5, 1.6, (7, 4),  $\pi/6$ }, as shown in Fig. 9(a). The initial velocities of the ellipses are respectively (0.005 m/s, 0) and (-0.005 m/s, 0), indicating they move towards each other along the x-axis. The mass density is set to 1000 kg/m<sup>3</sup>, and the normal contact stiffness is 1 N/m. This low stiffness is intentionally chosen to induce significant deformation and overlap during impact. Meanwhile, the impact duration can be prolonged so that the rotational motion of particles can be fully developed to make the contact more sophisticated. No gravity, damping and friction are considered. The time step is 7.927 s, and 53 steps are performed, resulting in an overall duration of approximately 420.1 s. Fig. 9(b)-(d) shows configurations of the two ellipses at various stages.

Fig. 10 displays the evolution of normalised translational, rotational and total kinetic energy of the systems throughout the impact. The conservation of total kinetic energy is evident, even during substantial overlap (see Fig. 9(c)). Notably, the rotational motion is fully developed, and over 40 % of kinetic energy is transferred into rotational energy by the end of the impact.

##### 4.2. Energy conservation verification with three ellipses in a confined box

The second example involves an elastic impact of three ellipses  $\xi_1$ : {2.0, 1.6, (5, 3), 0.0},  $\xi_2$ : {1.0, 0.8, (6.4, 5.8),  $\pi/6$ } and  $\xi_3$ : {1.8, 1.4, (4.5, 8.0),  $2\pi/3$ } within a confined square box of 10 m in length, as illustrated in Fig. 11(a). Initially, the top ellipse  $\xi_3$  has a downward velocity of (0, -0.1 m/s), while  $\xi_1$  and  $\xi_2$  are stationary. The mass density for all ellipses is 1 kg/m<sup>3</sup>, and the normal contact stiffness between both ellipses and ellipse-wall is set to 10<sup>5</sup>N/m, which is much bigger than that in the first example. Damping and friction are not considered in this system, but gravity is applied. The simulation uses a time step of 10<sup>-4</sup>s and runs for a total of 80,000 steps, resulting in an overall duration of 8 s.

Under the influence of gravity and the downward velocity of  $\xi_3$ , all ellipses begin to move and interact elastically with each other and the four walls of the box. Fig. 11(b)-(d) shows the positions of the ellipses at three-time instances during the simulation.

The evolution of the three energies of the system is depicted in Fig. 12, where the total energy includes the potential energy. Throughout the whole impact duration, over 100 individual elastic collisions occur between ellipse-ellipse and ellipse-wall. These results demonstrate that the total energy remains conserved.

##### 4.3. Illustrative simulation

After verifying the energy-conserving nature of the proposed framework, a more complex example is presented to demonstrate the method's applicability and robustness. Four dense ellipse packing assemblies are generated using the advancing front method (Xia et al., 2023, 2025). Coulomb's law is used to model friction in each contact, and the coefficient of friction is 0.3. To dissipate kinetic energy, a damping coefficient of 0.2 is applied in each contact. A linear viscous damping scheme is used and it can be envisioned as dashpots acting at contacts in both the normal and shear directions. The use of this damping effectively dissipates kinetic energy to obtain steady-state

solutions (Hart et al., 1988).

Then the ellipses settle under the action of gravity within a concave computational domain. The simplified contact detection and resolution proposed in Section 3.3 is applied to handle interactions between ellipses and convex/concave boundaries. Fig. 13 shows the packing configurations of the system at four different time instances. Due to friction, a stable packing with a distinct angle of repose forms by the end of each simulation. Additionally, the packing assemblies generated at the top of the domain tighten further at the bottom of the domain.

## 5. Discussion

### 5.1. Computational performance

To evaluate the accuracy and efficiency of the proposed framework,

a comparative analysis is performed. First, computational accuracy is assessed by approximating an ellipse with an inner polygon. As shown in Fig. 14, two ellipses ( $\xi_1$  and  $\xi_2$ ) with parameters matching those in Section 4.1 are discretized into convex polygons using adaptive sampling (Han et al., 2006). Table 4 summarizes the approximation errors of area and moment of inertia (MOI) for polygons with varying numbers of sides, with the area and MOI of the ellipses serving as the reference values. In the adaptive sampling method, the number of sides is controlled by the following criterion:

$$h/l < \tau \quad (23)$$

where  $h$  is the midpoint to segment distance;  $l$  is the segment length.

Four different tolerance levels ( $\tau = 0.038, 0.06, 0.1, 0.15$ ) are tested, resulting in the following side pairs of the two ellipses: (32, 28), (16, 20), (12, 12), and (8, 8). A large tolerance level leads to fewer sides and

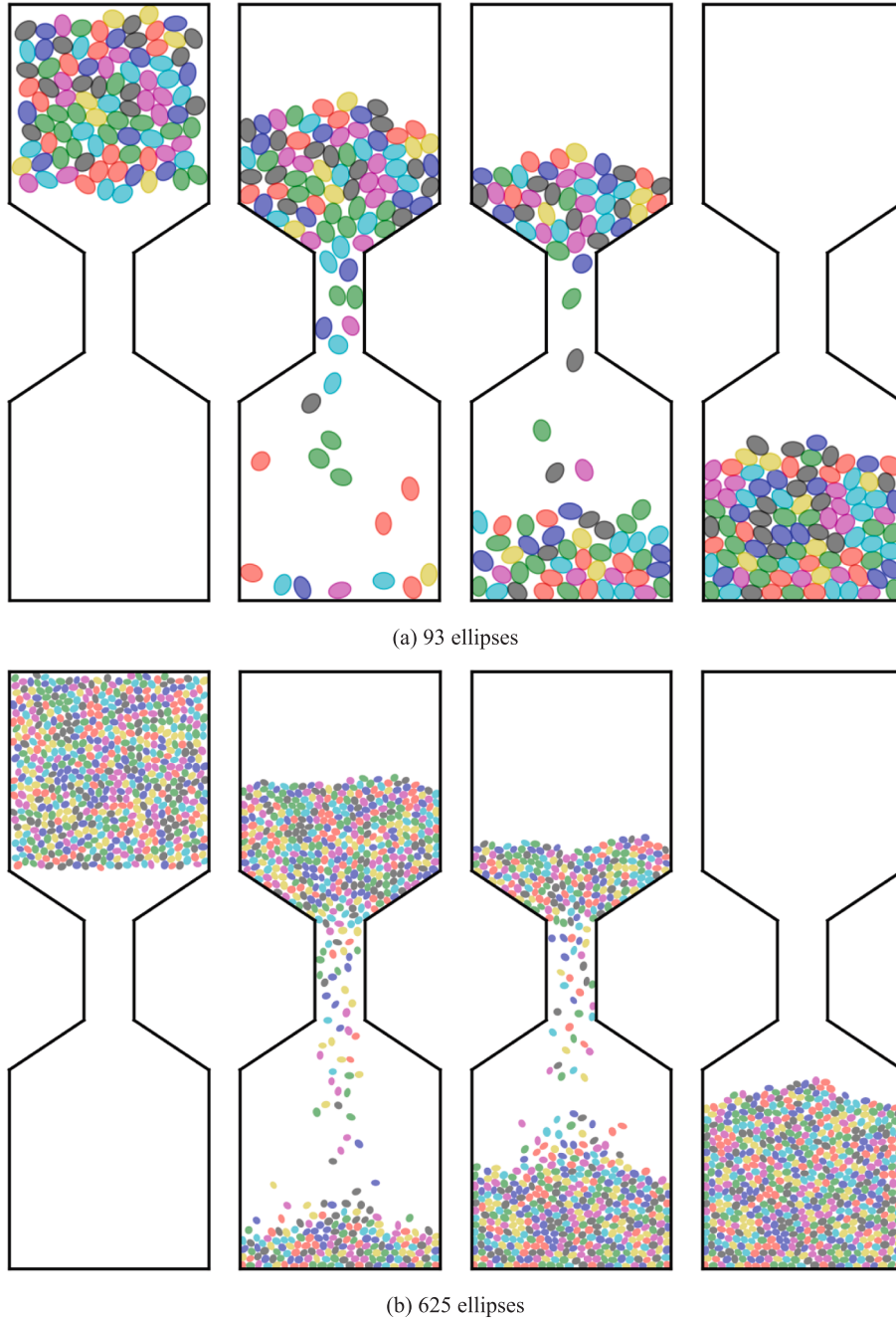
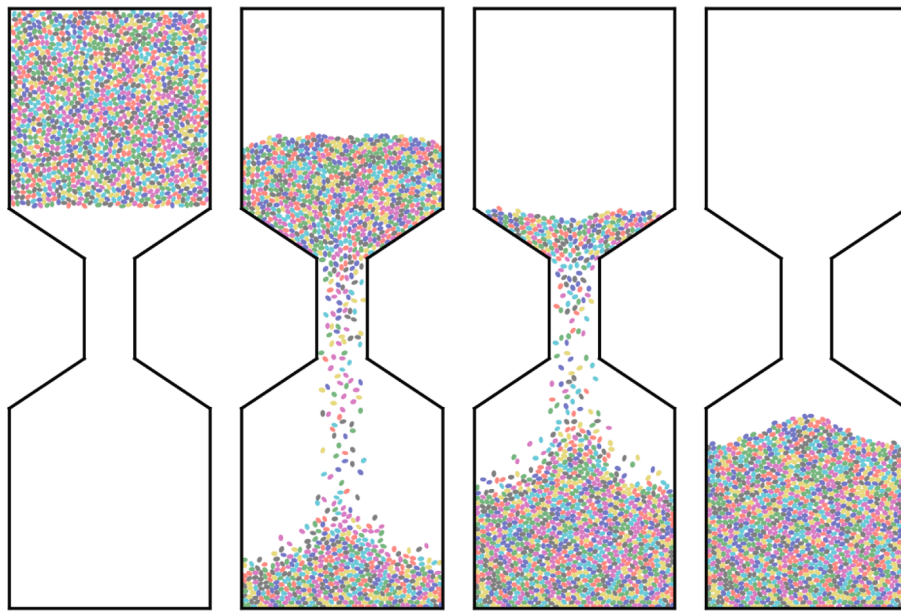
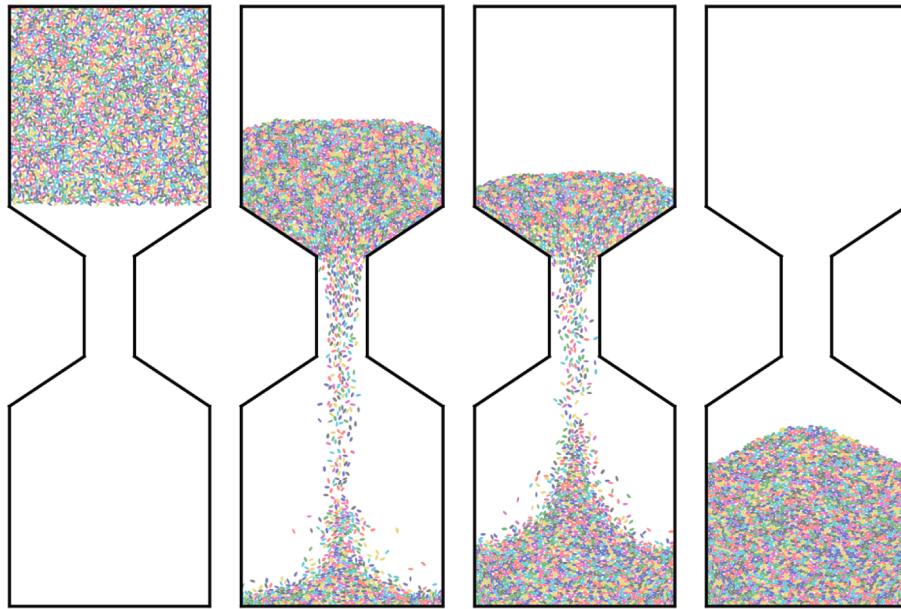


Fig. 13. Hopper filling with different ellipse numbers.



(c) 1357 ellipses



(d) 2677 ellipses

**Fig. 13.** (continued).

greater approximation errors. The smallest tolerance level ( $\tau = 0.038$ ) yields the most accurate approximations with the highest side counts, denoted as P32-P28. Meanwhile, ellipses are preferred over polygons due to their reduced geometric information and memory requirements.

To quantitatively evaluate how polygonal approximations affect particle mechanics during elastic impacts, these approximation polygons are used in modelling. Except for particle shapes, all computational parameters are the same as those in Section 4.1. Fig. 15 compares the time evolutions of three energies with five particle pairs, where the results of ellipses are used as reference. Notably, only P32-P28 achieves high accuracy, particularly in rotational motion (see Fig. 15(b)). Smooth

particles require sufficient polygon sides to achieve comparable accuracy to ellipses. In this example, P32-P28 matches the accuracy of ellipses, while P8-P8 exhibits the largest error as expected. Energy conservation during elastic impacts is maintained for all polygons, as shown in Fig. 15(c), where total energy remains unchanged before and after the collision. This occurs because the framework ensures energy conservation for both ellipses and polygons.

To assess the computational efficiency of the proposed approach, we measured the computational time for the hopper filling problem in Section 4.3 with varying numbers of ellipses. These ellipses were discretized into multi-sided polygons using three tolerance levels ( $\tau =$



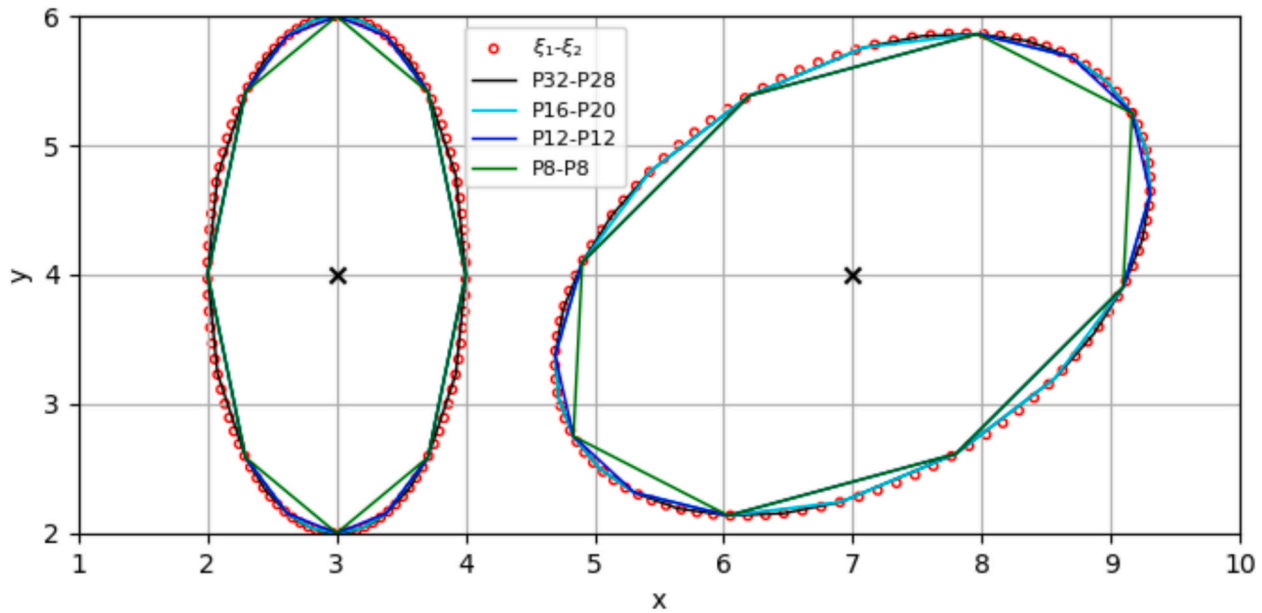


Fig. 14. Two ellipses and their inner polygonal approximation.

Table 4

Approximation errors of area and moment of inertia (MOI) for ellipses with different tolerance levels.

| Tolerance levels $\tau$ | No. sides of $\xi_1$ and $\xi_2$ | Area of $\xi_1$ |        | MOI of $\xi_1$ |         | Area of $\xi_2$ |        | MOI of $\xi_2$ |         |
|-------------------------|----------------------------------|-----------------|--------|----------------|---------|-----------------|--------|----------------|---------|
|                         |                                  | Value           | Error  | Value          | Error   | Value           | Error  | Value          | Error   |
| —                       | Ellipse                          | 6.283           | —      | 7853.982       | —       | 12.566          | —      | 27677.431      | —       |
| 0.038                   | 32, 28                           | 6.190           | 1.48 % | 7692.426       | 2.06 %  | 12.426          | 1.11 % | 27163.626      | 1.86 %  |
| 0.06                    | 16, 20                           | 5.920           | 5.77 % | 7250.642       | 7.68 %  | 12.306          | 2.07 % | 26645.311      | 3.73 %  |
| 0.1                     | 12, 12                           | 5.890           | 6.25 % | 7136.863       | 9.13 %  | 11.780          | 6.25 % | 24919.659      | 9.96 %  |
| 0.15                    | 8, 8                             | 5.657           | 9.96 % | 6380.712       | 18.76 % | 11.314          | 9.96 % | 22485.629      | 18.76 % |

0.038, 0.06, 0.1), employing GJK-EPA for contact detection and resolution. While GJK handles contact detection for both polygons and ellipses in DEM simulations, different strategies are used to compute contact features between contact pairs.

The comparison of computational cost for ellipses and their approximating polygons is shown in Fig. 16. As expected, computational cost increases with the number of particles. For the same number of particles, the proposed framework is most efficient when using ellipses, while polygons with the most sides (largest tolerance level  $\tau = 0.068$ ) are the least efficient. The simulated results show that the computational cost for polygons is about 3–4 times higher than those for ellipses, indicating that the proposed framework is very efficient. This good performance can be attributed to two key factors: (1) EPA-LM outperforms pure EPA in terms of computational efficiency; (2) reusing the previous contact angle as an initial guess in LM significantly reduces computational time.

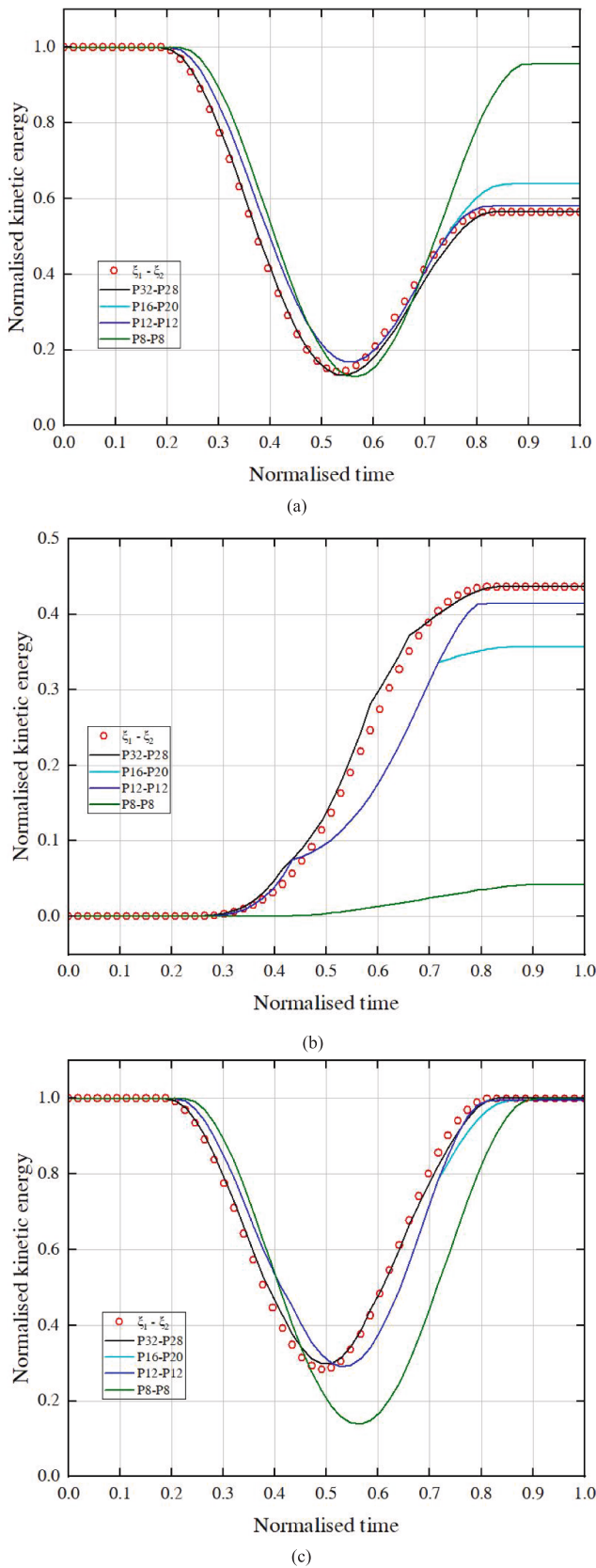
Additionally, since energy conservation is guaranteed among contacts, no numerical instability occurs in these examples, demonstrating the stability of the proposed framework.

## 5.2. Comparison with other solution strategies

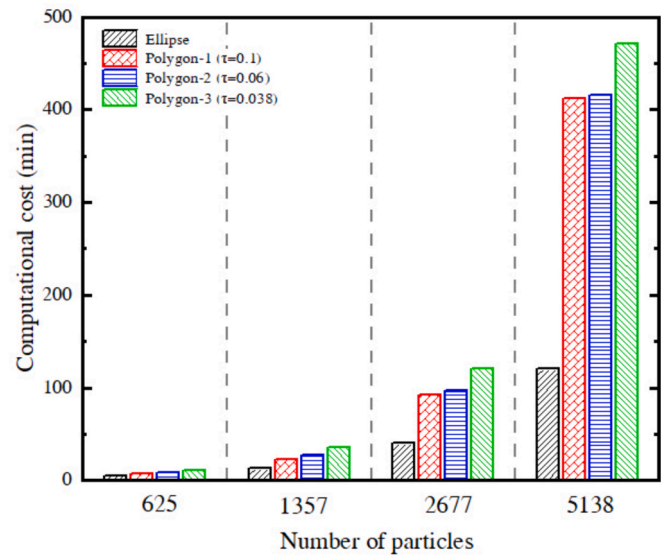
Previous study indicates that contact detection and resolution for ellipsoidal/poly-ellipsoidal particles may consume over 90 % of the total computation time in DEM simulations (Yan and Regueiro, 2019). Hence,

the algorithms for contact detection and resolution should be as efficient as possible to reduce computational costs. In the contact detection stage, GJK is used to identify contacts between ellipses, which is very efficient for overlap checking (Xia et al., 2023). In the contact resolution stage, the hybrid EPA-LM approach is employed for new contacts, while LM is employed for old contacts. Other solution strategies are also available for contact resolution, which are examined and analyzed below:

- (1) EPA. Although EPA can be used alone in the contact resolution stage, its computation cost is much higher than the hybrid EPA-LM approach, as discussed in Section 5.1. Additionally, numerical results from Feng and Tan (2020) demonstrate that EPA is less effective for curved and smooth surfaced shapes.
- (2) Hybrid LM and erosion-GJK. Besides the limitations of erosion-GJK outlined in the Introduction, contact overlaps in certain contact scenarios may exhibit very flat regions or multiple extrema. For flat regions, LM may fail or converge significantly slowly, and then erosion-GJK is used to resolve this issue (Zhao and Zhao, 2019, 2021). However, it is still challenging to determine accurate contact overlap with multiple extrema. For example, Fig. 22 in Kheradmand et al. (2022) illustrates a case with two local minima and two local maxima.
- (3) Newton-Raphson (NR) or EPA-NR. NR is the most used equation-solving procedure for determining two contact points ( $\mathbf{c}_1$  and  $\mathbf{c}_2$ ). However, it may diverge if an inappropriate initial guess is



**Fig. 15.** Time evolutions of normalised translational (a), rotational (b) and total (c) kinetic energies of five particle pairs.



**Fig. 16.** A comparison of computational costs for ellipses and their approximating polygons.

chosen. Notably, Eq. (11) yields two pairs of solutions, as outlined in Section 2.2.4. Although EPA can be employed to provide an initial estimate for  $c_1$  and  $c_2$ , the determination of two other unknowns (i.e.  $\mu$  and  $\nu$  in Table 1) remains challenging. Substituting  $c_1$  and  $c_2$  into Eq. (11) may produce solutions with imaginary numbers of  $\mu$  and  $\nu$ , leading to NR failure, as noted by Cleary et al. (1997).

Initially, we attempted to compare all four methods using Newton-Raphson (NR) iterations on realistic examples (e.g., hopper filling). However, only our framework and GJK/EPA demonstrated stable performance. The instability of the other methods stems from NR's sensitivity to initial guesses. As shown in Eq. (11), there are four unknown variables ( $c_1$ ,  $c_2$ ,  $\mu$  and  $\nu$  in Table 1) in the CN method. For existing contacts, the previous step's solution can be used as the initial guess for the present step. However, for new contacts, it is very difficult to determine these four variables, especially  $\mu$  and  $\nu$ . Additionally, for computational efficiency, we have provided direct comparisons between the present framework with EPA in Section 5.1.

## 6. Conclusions

This work provides a detailed comparison of four contact modelling approaches for ellipses: the intersection method, geometric potential method, midway method and common-normal method. It is found that only the common-normal method ensures energy conservation. Building on this analysis, this work proposes an accurate, stable and efficient contact detection framework based on the common-normal method. In the contact detection stage, the GJK algorithm is employed to identify contacts between ellipses. Once a contact is detected, a hybrid EPA-LM approach is utilized to compute contact features in a two-step contact resolution procedure. For contact detection and resolution between ellipses and convex/non-convex boundaries, a simplified procedure is proposed. The framework is verified through two numerical examples focused on energy conservation and then applied to modelling an illustration example with a large number of particles. Detailed computational performance and advantages of the proposed framework are presented. The proposed framework can be extended to ellipsoids/super-ellipsoids or other particles with convex smooth surfaces in 3D, which will be our future study.



## CRediT authorship contribution statement

**Ming Xia:** Writing – original draft, Resources, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Longjun Dong:** Writing – review & editing, Methodology, Conceptualization. **Feng-qiang Gong:** Methodology, Formal analysis, Conceptualization. **Jin Yu:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Tongming Qu:** Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Y.T. Feng:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Appendix A. The support point of shapes and Minkowski difference

In the present framework and GJK/EPA, support points of a shape and Minkowski difference of two geometric objects play a very important role. While the basic properties of support points of a shape are given as follows. For a given non-zero search direction vector  $\mathbf{n}$ , the support point (or function) of shape  $A$  along  $\mathbf{n}$ , denoted as  $\mathbf{p}_s(A, \mathbf{n})$ , is the furthest point of  $A$  along  $\mathbf{n}$ :

$$\mathbf{p}_s(A, \mathbf{n}) = \mathbf{p} \in A : \mathbf{p} \cdot \mathbf{n} \geq \mathbf{q} \cdot \mathbf{n}, \forall \mathbf{q} \in A \quad (\text{A1})$$

Fig. A1 depicts some examples of the support points. Table A1 gives some support points for different shapes.

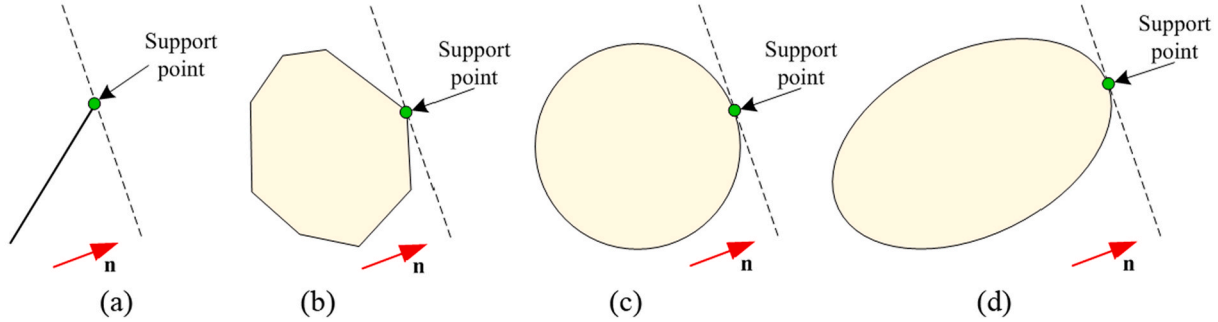


Fig. A1. Support points of some shapes: (a) segment; (b) polygon; (c) disc; (d) ellipse.

Table A1

Support points for different shapes.

| Shape   | Parameter                                                                               | Support point $\mathbf{p}_s(A, \mathbf{n})$ along the direction $\mathbf{n}$                                     |
|---------|-----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| Segment | vertices $\mathbf{v}_1, \mathbf{v}_2$                                                   | $\mathbf{v}_1$ for $\mathbf{v}_1 \cdot \mathbf{n} > \mathbf{v}_2 \cdot \mathbf{n}$ , or $\mathbf{v}_2$ otherwise |
| Polygon | vertices $\mathbf{v}_i$ ( $i = 1, \dots, k$ )                                           | $\mathbf{v}_i$ for which $\max\{\mathbf{v}_i \cdot \mathbf{n}\}$ ( $i = 1, \dots, k$ )                           |
| Disc    | radius $r$ and centre coordinates $\mathbf{c}_0$                                        | $\mathbf{c}_0 + r\mathbf{n}$                                                                                     |
| Ellipse | semi-major axes $a, b$ , central coordinates $\mathbf{c}_0$ and rotation angle $\theta$ | Eq. (14)                                                                                                         |

After obtaining two support points of  $A$  and  $B$ , the support point for the Minkowski difference of  $A$  and  $B$  along direction  $\mathbf{n}$  is.

$$\mathbf{p}_s(A \ominus B, \mathbf{n}) = \mathbf{p}_s(A, \mathbf{n}) + \mathbf{p}_s(-B, \mathbf{n}) = \mathbf{p}_s(A, \mathbf{n}) - \mathbf{p}_s(B, -\mathbf{n}) \quad (\text{A2})$$

Fig. A2 illustrates the Minkowski difference of a quadrilateral  $A$  and an ellipse  $B$ . Meanwhile, the support points of  $A$ ,  $B$ , and  $A \ominus B$  along search direction  $\mathbf{n}$  are also shown. The computational algorithm for explicit construction of  $A \ominus B$  can refer to Algorithms 1 and 2 in Xia et al. (2023). However, the explicit construction of Minkowski difference is computational inefficiency. In contrast, the proposed framework avoids explicit construction of the Minkowski difference of the two ellipses, significantly improving its computational efficiency.

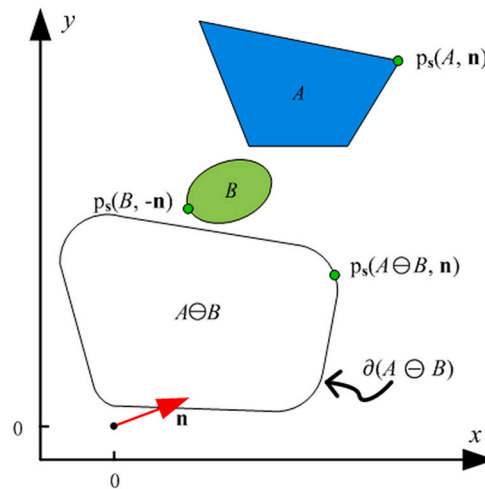


Fig. A2. The Minkowski difference of a quadrilateral  $A$  and an ellipse  $B$ .

## Data availability

Data will be made available on request.

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